Class 6 Science – Chapter 9: Methods of Separation in Everyday Life

इस अध्याय के सभी उपलब्ध पृष्ठों (118–127) के आधार पर नीचे दिए गए उत्तर Simple English में लिखे गए हैं। इनमें पुस्तक के प्रश्न, Think and Answer, Quick Check, गतिविधियों के उत्तर, प्रत्येक टॉपिक के महत्वपूर्ण प्रश्न-उत्तर तथा परीक्षा-अभ्यास शामिल हैं।

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1. Making Tea – Observe and Answer (Page 118)

Q1. Which object does Tamanna use to separate tea leaves from the mixture?

Answer: Tamanna uses a strainer to separate tea leaves from tea.

Q2. Can you name another object in the kitchen that is used for the separation of materials?

Answer: A sieve is another kitchen object used to separate materials, such as bran from flour.

Q3. What other things can be separated using this object?

Answer: A strainer can be used to separate seeds from juice and solid food particles from liquids.

Q4. Do you drink juice after separating the pulp of the fruits or not? Why?

Answer: It depends on personal preference. Some people separate the pulp to get clear juice, while others drink juice with the pulp because it contains fibre.

2. Topic: Mixtures and the Need for Separation

Q1. What is an impure substance?

Answer: An impure substance contains two or more substances mixed together. It does not have a fixed chemical composition.

Q2. What is a mixture?

Answer: A mixture is formed when two or more substances are mixed together and can be separated by suitable methods.

Q3. Why do we separate the components of a mixture?

Answer: We separate the components of a mixture to remove unwanted substances or to obtain useful substances separately.

Q4. Why should stones and weed seeds be removed from grains?

Answer: Stones and weed seeds are unwanted substances. They should be removed to make the grains clean and safe for use.

Q5. Why are grains cleaned before storage?

Answer: Grains are cleaned to remove stones, weed seeds, insects, and other impurities that may affect their quality.

Q6. Why are useful components separated from crude oil?

Answer: Useful components such as petrol, diesel, and kerosene are separated from crude oil so that they can be used for different purposes.

Q7. What is a homogeneous mixture?

Answer: A homogeneous mixture is a mixture in which the components are spread uniformly throughout it.

Q8. What is a heterogeneous mixture?

Answer: A heterogeneous mixture is a mixture in which the components are not spread uniformly throughout it.

Q9. Give one example each of a homogeneous and a heterogeneous mixture.

Answer:

- Homogeneous mixture: Salt dissolved in water.
- Heterogeneous mixture: Sand mixed with water.

3. Methods of Separation

Q1. What determines the method of separation used for a mixture?

Answer: The nature and physical properties of the components determine the method of separation.

Q2. Name some physical properties used to separate mixtures.

Answer: Particle size, weight, magnetic nature, state of matter, and solubility are some physical properties used for separation.

Q3. What is solubility?

Answer: Solubility is the ability of a substance to dissolve in another substance.

Q4. What are soluble substances?

Answer: Substances that dissolve in a liquid are called soluble substances. For example, sugar dissolves in water.

Q5. What are insoluble substances?

Answer: Substances that do not dissolve in a liquid are called insoluble substances. For example, sand does not dissolve in water.

Q6. Which organ of the human body purifies blood?

Answer: The kidneys filter wastes from the blood and help remove them from the body through urine.

4. Separation of Solids from Other Solids

A. Handpicking

Q1. What is handpicking?

Answer: Handpicking is a method of separation in which unwanted solid substances are removed from useful solids by hand.

Q2. When is handpicking used?

Answer: Handpicking is used when unwanted substances are easily visible, differ in size, shape, or colour, and are present in small quantities.

Q3. Give two examples of handpicking.

Answer:

1. Removing stones from rice.
2. Removing unwanted particles from pulses.

Q4. Why is handpicking not suitable for separating very large quantities of tiny particles?

Answer: It is slow and difficult when the mixture contains many tiny particles.

B. Sieving

Q1. What is sieving?

Answer: Sieving is a method of separation in which a sieve is used to separate particles of different sizes.

Q2. How does a sieve work?

Answer: Small particles pass through the holes of the sieve, while larger particles remain on it.

Q3. Give two examples of sieving.

Answer:

1. Separating bran or larger particles from flour.
2. Separating stones and pebbles from sand at a construction site.

Q4. Why are sieves made with different hole sizes?

Answer: Different hole sizes are needed to separate particles of different sizes.

C. Threshing

Q1. What is threshing?

Answer: Threshing is the process of separating grains from the stalks of harvested crops.

Q2. How is threshing done traditionally?

Answer: Traditionally, farmers beat the stalks against a hard surface or with a stick, or allow animals such as bullocks to trample the stalks.

Q3. How is threshing done today?

Answer: Threshing can also be done using machines.

Q4. Why is threshing important in agriculture?

Answer: It separates grains from the stalks after harvesting, making the grains easier to collect and use.

D. Winnowing

Q1. What is winnowing?

Answer: Winnowing is a method in which wind is used to separate heavier components from lighter components of a mixture.

Q2. How do farmers separate husk from grains by winnowing?

Answer: Farmers drop the mixture from a height. The lighter husk is carried away by the wind, while the heavier grains fall nearby.

Q3. Why is husk separated from grains by winnowing?

Answer: Husk is lighter than grains, so wind can carry it away more easily.

Q4. What is the difference between threshing and winnowing?

Threshing	Winnowing
Separates grains from stalks.	Separates lighter husk from heavier grains.
Uses beating, trampling, or machines.	Uses wind or moving air.
Done after harvesting.	Often done after threshing.

E. Magnetic Separation

Q1. What is magnetic separation?

Answer: Magnetic separation is a method in which a magnet is used to separate magnetic materials from non-magnetic materials.

Q2. How can iron filings be separated from sand?

Answer: Move a magnet over the mixture. The iron filings will be attracted to the magnet, while the sand will remain behind.

Q3. Name some materials attracted by a magnet.

Answer: Iron, nickel, and cobalt are examples of materials attracted by magnets.

Q4. Can a magnet separate sand from iron filings? Why?

Answer: Yes. Iron filings are attracted by a magnet, but sand is not.

Q5. How can scrap iron be separated from a heap of waste material?

Answer: A magnet or magnetic separator can be used to attract and remove the iron pieces from the waste.

5. Separation of Insoluble Solids from Liquids

A. Sedimentation

Q1. What is sedimentation?

Answer: Sedimentation is the process in which heavier insoluble particles settle at the bottom of a liquid when the mixture is left undisturbed.

Q2. What are sediments?

Answer: The solid particles that settle at the bottom of a liquid are called sediments.

Q3. Give an example of sedimentation.

Answer: When muddy water is left undisturbed, mud settles at the bottom.

B. Decantation

Q1. What is decantation?

Answer: Decantation is the process of carefully pouring out the clear liquid without disturbing the settled solid particles.

Q2. Why are sedimentation and decantation often used together?

Answer: Sedimentation allows the heavier particles to settle, and decantation allows the clear liquid to be poured off.

Q3. How can sand be separated from water by sedimentation and decantation?

Answer:

1. Mix sand with water.
2. Leave the mixture undisturbed.
3. Allow the sand to settle at the bottom.
4. Carefully pour the clear water into another container.

Q4. Why are rice and pulses sometimes soaked and washed before cooking?

Answer: Soaking and washing help loosen and remove dirt and some unwanted particles.

C. Filtration

Q1. What is filtration?

Answer: Filtration is a method of separating an insoluble solid from a liquid by passing the mixture through a filter.

Q2. What is a filter?

Answer: A filter is a material with small pores that allows liquid to pass through but holds back solid particles that are too large to pass through.

Q3. What is the residue?

Answer: The solid left behind on the filter is called the residue.

Q4. What is the filtrate?

Answer: The liquid that passes through the filter is called the filtrate.

Q5. Give three examples of filtration.

Answer:

1. Separating tea leaves from tea.
2. Separating mud from muddy water.
3. Separating pulp from fruit juice.

Q6. Can ordinary filtration remove all dissolved substances from water?

Answer: No. Ordinary filtration removes suitable insoluble particles, but dissolved substances such as salt can pass through with the water.

Q7. Name some materials that can be used as filters other than filter paper.

Answer: A clean cloth, cotton wool, and suitable mesh materials can be used as filters, depending on the size of the particles.

Q8. What is loading?

Answer: Loading is a method in which alum is added to water to help very fine suspended particles become heavier and settle more quickly.

Q9. What is the difference between sedimentation and filtration?

Sedimentation	Filtration
Particles settle under gravity.	A filter holds back suitable solid particles.
The mixture is left undisturbed.	The mixture is passed through a filter.
Example: Mud settling in water.	Example: Tea leaves separated by a strainer.

6. Quick Check – Fill in the Blanks (Page 123)

Q1. _____ are substances that cannot be further broken down into simpler substances by physical or chemical methods.

Answer: Elements.

Q2. Mg is the symbol of _____.

Answer: Magnesium.

Q3. In a _____ mixture, the constituents are spread uniformly throughout the mixture.

Answer: Homogeneous.

Q4. _____ is the property of a substance to get dissolved in another substance.

Answer: Solubility.

Q5. In the process of _____, bullocks are allowed to trample the stalks.

Answer: Threshing.

Q6. A mixture of iron and sand can be separated by _____ separation.

Answer: Magnetic.

7. Separation of Soluble Solids from Liquids

A. Solute, Solvent, and Solution

Q1. What is a solute?

Answer: A solute is a substance that dissolves in another substance. For example, salt is the solute in salt water.

Q2. What is a solvent?

Answer: A solvent is the substance in which a solute dissolves. Water is the solvent in salt water.

Q3. What is a solution?

Answer: A solution is a mixture formed when a solute dissolves in a solvent.

Q4. What is solubility?

Answer: Solubility is the ability of a substance to dissolve in another substance.

B. Evaporation

Q1. How can salt be separated from salt water?

Answer: Heat the salt water in a suitable dish. The water evaporates, leaving the salt behind.

Q2. What is evaporation used for in the separation of mixtures?

Answer: Evaporation is used to separate a dissolved solid from a liquid, such as salt from salt water.

Q3. How is common salt obtained from seawater?

Answer: Seawater is collected in shallow ponds and exposed to sunlight. The water evaporates, leaving salt behind. The salt is then further purified.

Q4. Can evaporation recover both salt and water from salt water?

Answer: Evaporation alone leaves the salt behind, but the water is lost as vapour. To recover the water as well, the vapour must be condensed and collected.

Q5. What is the difference between filtration and evaporation?

Answer: Filtration separates insoluble solids from liquids. Evaporation separates a dissolved solid from a solution by changing the liquid into vapour.

C. Condensation

Q1. What is condensation in the separation of mixtures?

Answer: Condensation is the process of changing vapour into liquid by cooling it.

Q2. How can water be recovered from a salt solution?

Answer: Heat the salt solution to form water vapour and then cool the vapour so that it condenses into liquid water. The salt remains behind.

Q3. What is distillation?

Answer: Distillation is a method of separating components of a liquid mixture using evaporation followed by condensation.

Q4. Why is condensation useful after evaporation?

Answer: Condensation helps recover the liquid that has changed into vapour.

D. Churning

Q1. What is churning?

Answer: Churning is the process of separating butter from curd or milk by continuous stirring or agitation.

Q2. Which tool is traditionally used for churning?

Answer: A wooden churner, also called a mathani, is traditionally used.

Q3. How is churning done today?

Answer: Electric and mechanical churners can be used.

Q4. Is churning the same as filtration?

Answer: No. Churning separates butter through agitation, while filtration separates suitable insoluble solids from liquids using a filter.

8. Separation Using More Than One Method

Q1. Why may we need more than one method to separate a mixture?

Answer: A mixture may contain several components with different properties. One method may not be enough to separate all the components.

Q2. How can a mixture of salt, water, and chalk powder be separated?

Answer:

1. Leave the mixture undisturbed so that the chalk powder settles.
2. Use decantation or filtration to separate the chalk from the salt solution.
3. Heat the salt solution to evaporate the water.
4. The salt remains in the dish. If water is also required, collect and condense the vapour.

Q3. How can a mixture of sand and salt be separated?

Answer:

1. Add water to the mixture and stir. The salt dissolves, but the sand does not.
2. Allow the sand to settle.
3. Decant the salt solution or filter the mixture.
4. Heat the salt solution to evaporate the water.
5. Collect the salt left behind.

Q4. How can pebbles, sand, sugar, and iron filings mixed with water be separated?

Answer: The methods can be applied in the following order:

1. Filtration: Separate the insoluble solids from the sugar solution.
2. Handpicking or sieving: Separate the larger pebbles from the smaller solids.
3. Magnetic separation: Separate iron filings from the remaining solids.
4. Sieving: Separate sand from larger pebbles, if their particle sizes differ sufficiently.
5. Evaporation: Separate sugar from the sugar solution.

Note: The exact order of the solid-separation steps can vary depending on the sizes of the particles. Condensation can be used to recover water from the solution if required.

9. Activities and Experiments

Activity 1: Separating Mud from Water

Aim: To separate mud from water by filtration.

Materials required: Beaker, funnel, filter paper, flask, mud, and water.

Procedure:

1. Mix mud with water.
2. Fold the filter paper into a cone and place it in the funnel.
3. Pour the mixture through the filter paper.
4. Collect the liquid in a flask.

Observation: Mud particles remain on the filter paper, and water passes through.

Conclusion: Filtration separates suitable insoluble particles from a liquid.

Activity 2: Separating Salt from Water

Aim: To separate salt from salt water by evaporation.

Materials required: Salt, water, beaker, China dish, tripod stand, and a heat source.

Procedure:

1. Dissolve salt in water.
2. Transfer the solution to a suitable dish.
3. Heat the solution carefully under adult or teacher supervision.
4. Allow the water to evaporate.

Observation: Salt remains in the dish after the water evaporates.

Conclusion: Evaporation can separate a dissolved solid from a liquid.

Activity 3: Separating Sand and Salt

Aim: To separate a mixture of sand and salt.

Procedure:

1. Add water to the sand and salt mixture.
2. Stir until the salt dissolves.
3. Allow the sand to settle.
4. Decant the salt solution or filter it.
5. Heat the salt solution to evaporate the water.

Observation: Sand is separated first, and salt remains after evaporation.

Conclusion: A combination of separation methods can separate the components of a mixture.

10. Additional Important Questions and Answers

Q1. Which method is used to separate stones from rice?

Answer: Handpicking.

Q2. Which method is used to separate flour from larger particles?

Answer: Sieving.

Q3. Which method separates grains from stalks?

Answer: Threshing.

Q4. Which method separates husk from grains?

Answer: Winnowing.

Q5. Which method separates iron filings from sand?

Answer: Magnetic separation.

Q6. Which method separates mud from water?

Answer: Filtration, or sedimentation followed by decantation.

Q7. Which method is used to obtain salt from seawater?

Answer: Evaporation.

Q8. Which method is used to recover liquid water from salt water?

Answer: Distillation, which involves evaporation followed by condensation.

Q9. Why can sand and salt be separated by dissolving the mixture in water?

Answer: Salt dissolves in water, while sand does not. This difference in solubility allows them to be separated.

Q10. Why does a magnet not separate sand from salt?

Answer: Neither sand nor ordinary salt is attracted by a magnet, so magnetic separation is not suitable for separating them from each other.

Q11. Why is sieving not suitable for separating dissolved salt from water?

Answer: Dissolved salt particles pass through the sieve along with the water. Sieving separates particles mainly on the basis of size.

Q12. Why is filtration not enough to obtain pure water from salt water?

Answer: Dissolved salt passes through ordinary filter paper with the water. Filtration cannot remove dissolved salt.

Q13. What happens when too much sugar is added to water?

Answer: At a given temperature, water can dissolve only a limited amount of sugar. Once the solution becomes saturated, additional sugar remains undissolved and settles at the bottom.

Q14. What is a saturated solution?

Answer: A saturated solution is a solution that cannot dissolve any more of a particular solute under the given conditions.

Q15. How can we separate butter from curd?

Answer: Butter can be separated from curd by churning.

Q16. Why should heating experiments be performed under supervision?

Answer: Heating can cause burns or accidents. Experiments should be performed carefully under the supervision of a teacher or an adult.

11. Multiple-Choice Questions (MCQs)

Chapter Quiz

0/12 answered

1. Which method is used to remove stones from rice?

A. Filtration
B. Handpicking
C. Evaporation
D. Condensation

2. Which method separates grains from stalks?

A. Winnowing
B. Sieving
C. Threshing
D. Churning

3. Which method uses wind to separate husk from grains?

A. Filtration
B. Winnowing
C. Sedimentation
D. Handpicking

4. Which method separates iron filings from sand?

A. Evaporation
B. Magnetic separation
C. Decantation
D. Sieving

5. What settles at the bottom during sedimentation?

A. Solvent
B. Sediment
C. Vapour
D. Filtrate

6. What is the liquid that passes through a filter called?

A. Residue
B. Sediment
C. Filtrate
D. Solute

7. Which method is used to separate tea leaves from tea?

A. Filtration
B. Threshing
C. Winnowing
D. Churning

8. Which method separates salt from salt water?

A. Handpicking
B. Evaporation
C. Winnowing
D. Sieving

9. What is the substance that dissolves in a solvent called?

A. Solute
B. Residue
C. Filter
D. Sediment

10. What is the process of separating butter from curd called?

A. Decantation
B. Churning
C. Filtration
D. Condensation

11. Which method can recover water from salt solution?

A. Handpicking
B. Sieving
C. Distillation
D. Threshing

12. A mixture in which components are spread uniformly is called:

A. Heterogeneous mixture
B. Homogeneous mixture
C. Suspension only
D. Pure element
Check Answers

12. Fill in the Blanks

1. Removing unwanted particles by hand is called handpicking.

2. Separating grains from stalks is called threshing.
3. Separating husk from grains using wind is called winnowing.
4. A magnet attracts iron filings.
5. The settling of heavier particles is called sedimentation.
6. Pouring out clear liquid without disturbing the sediment is called decantation.
7. Separating insoluble solids from liquids using a filter is called filtration.
8. The substance that dissolves in a solvent is called the solute.
9. The substance in which a solute dissolves is called the solvent.
10. Separating salt from salt water by heating is called evaporation.
11. Changing vapour into liquid is called condensation.
12. Separating butter from curd is called churning.

13. True or False

1. Handpicking is useful when unwanted substances are easily visible. — True
2. Sieving separates particles mainly on the basis of size. — True
3. Winnowing uses a magnet to separate husk from grains. — False
4. Threshing separates grains from stalks. — True
5. Sand dissolves completely in water. — False
6. Sedimentation allows heavier particles to settle at the bottom. — True
7. Filtration can remove all dissolved salt from water. — False
8. Evaporation can be used to obtain salt from salt water. — True
9. Condensation changes liquid water into water vapour. — False
10. Churning is used to separate butter from curd. — True
11. A saturated solution cannot dissolve more solute under the given conditions. — True
12. More than one method may be needed to separate a complex mixture. — True

14. Quick Revision Table

Method	What it separates	Example
Handpicking	Visible unwanted solids	Stones from rice
Sieving	Solids of different sizes	Larger particles from flour
Threshing	Grains from stalks	Harvested crops
Winnowing	Lighter and heavier solids	Husk from grains
Magnetic separation	Magnetic and non-magnetic materials	Iron filings from sand
Sedimentation	Heavier insoluble solids from liquid	Mud settling in water
Decantation	Clear liquid from settled solids	Pouring off water above sand
Filtration	Insoluble solids from liquids	Tea leaves from tea
Evaporation	Dissolved solid from liquid	Salt from salt water
Condensation	Vapour into liquid	Recovering water vapour
Distillation	Components using evaporation and condensation	Recovering water from salt solution

Churning	Butter from curd or milk	Making butter

Important: The uploaded PDF contains pages 118–127, including the chapter's summary diagram and activities. It does not show a separate end-of-chapter exercise page. Therefore, the answers above cover all questions visible in the uploaded pages and add topic-wise practice questions based on the chapter's content.

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